

Adaptive real-time interactive rendering of gigantic multi-resolution models

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Figure 1: View-dependent rendering of the Egyptian wall paint. The full resolution model contains 45 million triangles and is inspected at over 60 fps on a modest PC platform. The main images present the mesh rendered with Phong shading using texture mapping on a 1920x1080 window, while the small inset figures depict the adaptive mesh structure with a different color for different LOD. The rightmost image also shows the comparison of adaptive LOD rendering and direct LOD rendering.

ABSTRACT

Real-time visualization of large-scale surface models is still a challenging problem. When using multiple levels of details (LOD), the main issues are popping between levels and/or cracks between level parts. We present a scheme (both mesh preprocessing and real-time rendering), which avoids both of these issues. Vertex interpolation between child and parent is used to achieve crack and popping free rendering. We implemented and tested our method on a modest laptop PC, and could render scanned models of multiples tens of million triangles at optimal visual quality and interactive frame rate.

KEYWORDS

Real-time Rendering, 3D visualization, Multi-resolution Modeling

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1 INTRODUCTION

Real-time interactive visualization of massive 3D-scale scene data that maintains realism and frame rate requirements remains a challenging problem. One of the strategies is to generate multi-resolution models containing different LODs. In practice, it is complicated to adapt the resolution dynamically without introducing rendering artifacts such as popping during the LOD switching or cracks at the boundary between different LOD parts, because of the different LOD resolutions.

One of the solutions is based on progressive mesh, such as [Derzapf and Guthe 2012], [Derzapf et al. 2010] and [Hu et al. 2010]. These methods need to index vertices of boundaries in every state. This will prevent cracks perfectly, but the computation time and complexity are expensive. At the same time, there is no need to render every part of the model in fine granularity.

[Cignoni et al. 2005] and [Yoon et al. 2005] build the boundary constraints and cluster dependencies during the hierarchical simplification. These methods avoid cracks and reduce the popping artifacts by controlling the screen space error associated with a patch.

In this paper, we present a preprocessing and real-time visualization scheme which can avoid LOD popping and cracks. Our preprocessing implementation (starting from standard mesh file formats) achieves competitive levels in terms of treated triangles per second per core. We conducted test experiments on real large-scale archaeological data (such as murals). The method can realize real-time interactive rendering, which can facilitate further data analysis for other researchers through high-detail mesh data.

2 THE OVERVIEW OF OUR METHOD

Our method includes multi-resolution model construction and real-time view-dependent rendering. The overview of the hierarchical LOD construction process is shown in Figure 2. The algorithm of real-time view-dependent rendering is shown in Figure 3.

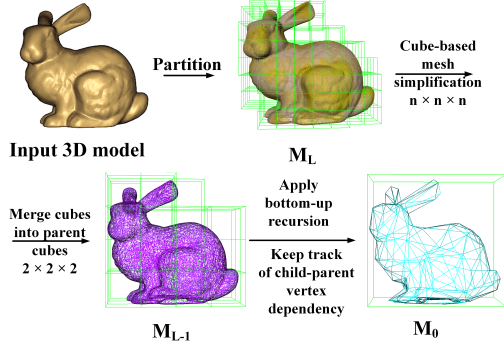


Figure 2: The overview of hierarchical multi-resolution model construction, where L is the highest LOD level. The cube-based simplification is based on edge collapse, and we simplify $n \times n \times n$ cubes at one time, usually, where $n > 2$. We merge $2 \times 2 \times 2$ simplified cubes to build parent cubes, which will build into a new level of LOD (lower resolution).

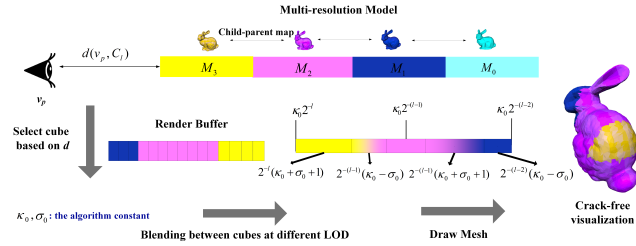


Figure 3: The overview of view-dependent rendering algorithm. The input is a multi-resolution model whose data will be uploaded to the GPU if the cube is within the visible distance and has passed frustum culling.

3 RESULTS

All the tests were done on a laptop machine equipped with an Intel i7-7700HQ CPU (16GB) and Nvidia GeForce 1060 Mobile (6GB). The test result is based on 4 models: Lucy, Egyptian tomb, Egyptian paint, and Egyptian temple wall paint. We implemented multi-resolution model construction and viewpoint-based visualization in turn.

3.1 Multi-Resolution Model Construction

Table 1 shows the number of triangles (n_t) and vertices (n_v) in both original mesh M_L and coarsest mesh M_0 , where $L = 6$, and the building time for multi-resolution model construction under 4 CPU cores.

Table 1: Quantitative results of multi-resolution model construction

Object	M_L		M_0		Time (s)
	n_v	n_t	n_v	n_t	
Tomb	998758	1995376	138824	115565	2.16
Lucy	14027872	28055742	3638883	7277732	21.62
Paint	22548641	45056702	2626808	2651781	51.52
Temple	30001155	60000000	2947716	2707987	72.53

3.2 Adaptive real-time rendering

By controlling the movement of the viewpoint to the model from a distance, the real-time rendering and full-resolution model rendering performance of the 4 models is shown in Figure 4.

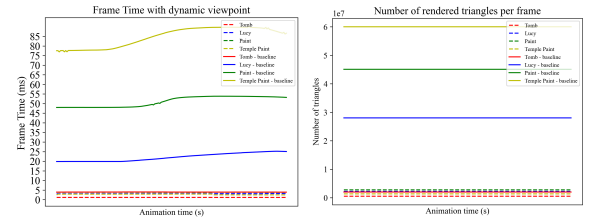


Figure 4: Rendering performances of 4 models with a dynamic viewpoint.

4 CONCLUSION

We present an efficient scheme for multi-resolution model construction and interactive view-dependent visualization of massive surface models on a general graphics platform. This scheme allows for the utilization of rendering different sizes of models on different graphics platforms.

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REFERENCES

- P. Cignoni, F. Ganovelli, E. Gobbetti, F. Marton, F. Ponchio, and R. Scopigno. 2005. Batched multi triangulation. In *VIS 05. IEEE Visualization, 2005*. 207–214. <https://doi.org/10.1109/VISUAL.2005.1532797>
- Evgenij Derzpf and Michael Guthe. 2012. Dependency-free parallel progressive meshes. In *Computer Graphics Forum*, Vol. 31. Wiley Online Library, 2288–2302. <https://doi.org/10.1109/VISUAL.2005.1532797>
- Evgenij Derzpf, Nicolas Menzel, and Michael Guthe. 2010. Parallel view-dependent out-of-core progressive meshes. *Realismus der Echtzeitgrafik* 61 (2010).
- Liang Hu, Pedro V. Sander, and Hugues Hoppe. 2010. Parallel View-Dependent Level-of-Detail Control. *IEEE Transactions on Visualization and Computer Graphics* 16, 5 (2010), 718–728. <https://doi.org/10.1109/TVCG.2009.101>
- S.-E. Yoon, B. Salomon, R. Gayle, and D. Manocha. 2005. Quick-VDR: out-of-core view-dependent rendering of gigantic models. *IEEE Transactions on Visualization and Computer Graphics* 11, 4 (2005), 369–382. <https://doi.org/10.1109/TVCG.2005.64>